

```
def compute_phi(network, state):  
    # 1. Генерируем все partitions  
    partitions =  
generate_all_partitions(network.nodes)  
  
    # 2. Для каждой partition вычисляем:  
    phi_candidates = []  
    for partition in partitions:  
        I_whole = mutual_information(network,  
state) # Информация целого  
        I_parts = sum(mutual_information(part,  
state)  
                        for part in partition)  
        phi_candidates.append(I_whole - I_parts)  
  
    # 3. Берем minimum по MIP (Minimum  
Information Partition)  
    return min(phi_candidates)
```